SSAD course

Prof NACER **University year: 2026-2027**

U.S.T.H.B

Practical work no1 (Free Programming)

Realizing an encryption framework

***Part 1*: An Interface containing (Encryption and Decryption, Password, Attacks, Authenticity or Integrity):**

***Part 2*: Encryption/ Decryption**

An entity A sends a message to an entity B, and the entity B decrypts the message.

The encryption approach can be:

- **Hill encryption approach;**
- **PlayFaire approach;**
- **Caesar encryption: shift Right-Left/ Left-Right by n rank regarding the letters of the Latin alphabet.**

Note: Deal with the security flaws of each algorithm. You have to realize the 3 algorithms.

***Part 3:* Passwords**

- **A password composed by 3 characters** such as each character can take the value 0 and 1
- **A password composed by 4 characters** such as each character can take the values 1,2,3,4.

- **A password** such as each character of the password can take the values a.. z,A..Z,0..9,+,*, etc.

- **Hacking methods**:
  - **Attacks** in the 03 cited use case with minimal time. A Dictionary attack and Brute Force attack.
- **Password protection** : Propose the solution to deal with the cited attacks

Part 4: Authenticity / Integrity

Realize one objective

The requested work:

- To share the different tasks of the project between the members
- Performance evaluation and tests
- Demonstration and a report of the forth parts